



King Abdul Aziz University

Faculty of Computing & Information Technology Computer Science Department

Final Exam (Tuesday, May 17, 2016)
Course: CPCS204 (Data Structures)
Term: 2nd Semester (2016), Time: 2 hours

Student ID: _____ Section: _____

Student Name: _____

Instructor:

Part I (Multiple Choice/Short Answer – 2 pts each)

CLO-9

1. When a Stack is implemented using an array, the **isFull()** method returns true when

- a) `top == -1`
- b) `top == maxSize`
- c) `top == maxSize - 1`
- d) `top == null`

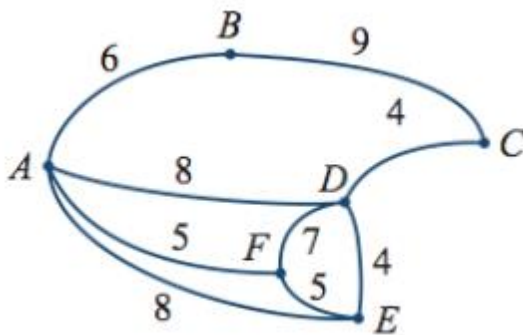
CLO-6

2. How many linked lists are used to represent a graph with n nodes and m edges, when using an adjacency- list representation?

- a) m
- b) n
- c) $n*m$
- d) $n+m$

CLO-4

3. Perform a **Depth-First** traversal of the weighted graph shown below, **starting with vertex A**. Select the smallest-weight edge first, when accessing neighbors of a vertex. What is the order in which the vertices are visited?



- a) AFDECB
- b) AFEDCB
- c) ABCDEF
- d) ADEFCB

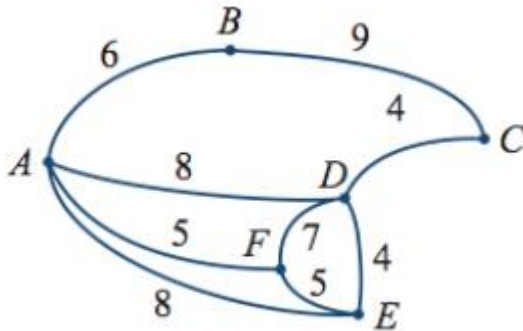
CLO-6

4. A complete graph with 5 vertices has how many edges

- a) 5
- b) 8
- c) 10
- d) 11

CLO-4

5. Perform a Breadth-First traversal of the weighted graph shown below, **starting with vertex A**. Select the smallest-weight edge first, and then alphabetical (in case of a tie), when accessing neighbors of a vertex. What is the order in which the vertices are visited?



- a) AFBDEC
- b) ABFEDC
- c) AFBEDC
- d) AFEBDC

CLO-4

6. A sorted array has 16 items. Using binary search, what is the maximum number of comparisons required to find a value in the array?

- a) 1
- b) 4
- c) 8
- d) 15

CLO-13

7. What are the three main steps of Merge Sort (give them in correct order)?

CLO-13

8. What are the three main steps of Quick Sort (give them in correct order)?

CLO-2

1) (10 points) Order Notation. Give the Big-O running times of the following operations in terms of n :

a) Finding a value in a Hash Table whose hash function evenly distributes all items:

f) Inserting or deleting exactly one element from a Heap (Priority Queue):

b) Sorting n elements using Selection Sort (best case):

g) Sorting n elements using Insertion Sort (worst case):

c) Delete all nodes from a Binary Heap:

h) Pop and print all elements in a stack.

d) Given an Adjacency Matrix, determine if an edge is between two specific nodes:

i) Given an Adjacency List, determine if an edge is between two specific nodes:

e) Searching for a value in a sorted array of n items:

j) Finding the largest value in a linked-list.

CLO-1

2) (8 points) Comparison of Data Structures. Answer the following questions:

- What are two disadvantages an array?
- In what way is a linked list better than an array?
- Why is a Binary Search Tree better than both arrays and linked-lists?
- Why is a Hash Table even better?

CLO-7 3) (10 points) **Linked Lists.** Write a method that operates on an existing linked list of integers. Your method should insert a new node at the very end of the list. The value of this new node should be 5 times the value of the last node.

```
public void addNewNode5x() {
```

```
}
```

CLO-16 4) (4 points) **Heaps.** Insert the following values into a MIN-Heap:

4 3 2 1 5 6 7 8

Solution:

CLO-8 5) (10 points) **Recursion.** Write a RECURSIVE method, `printTriangle`, that prints a triangle of asterisks whose length and height are of size, n . For example, if `printTriangle(5)` is called from main, the following will print to the screen:

```
* * * * *
* * * *
* * *
* *
*
```

*Note: you will **not** get points unless you use recursion!

HINT: Although you will use recursion to solve this, there can be a `for` loop inside the recursive function that you will write.

```
public void printTriangle(int n) {
```

```
}
```

CLO-13 6) (7 points) **Sorting.** Show the contents of the array below being sorted in ***ascending*** (from smallest to largest) order using **Selection Sort** at the end of each loop iteration.

Index	0	1	2	3	4	5	6	7
Initial Array:	55	45	65	75	70	50	40	60
Loop 1								
Loop 2								
Loop 3								
Loop 4								
Loop 5								
Loop 6								
Loop 7								
Sorted Array:	40	45	50	55	60	65	70	75

CLO-12 7) (10 points) **Binary Tree Code.** Suppose you have an existing binary tree, and each node stores an integer value. Write an appropriate method to count the number of even nodes in the tree and returns that count.

```
public int cEN(intBSTnode p) {
```

```
}
```

CLO-14 8) (9 points) **Sorting.** Show the contents of the array below, after each call to the Merge method occurs, in the process of running Merge-Sort on said array.

Index	0	1	2	3	4	5	6	7
Initial Array:	70	50	80	20	40	60	30	10
after 1 st Merge:								
after 2 nd Merge:								
after 3 rd Merge:								
after 4 th Merge:								
after 5 th Merge:								
after 6 th Merge:								
Sorted Array:	10	20	30	40	50	60	70	80

CLO-14 9) (6 points) **Sorting.** Show the result of running **Partition** (from Quick sort) on the array below using the leftmost element as the pivot element. Show what the array looks like after each swap. And then show the array after the partition (**remembering the final swap**).

Index	0	1	2	3	4	5	6	7
Initial Array:	50	80	20	90	60	30	40	70
after 1 st Swap:								
after 2 nd Swap:								
After Partition:								

CLO-15 10) (10 points) **Hash Tables.** Insert the following numbers (in the order that they are shown.....from left to right) into a hash table with an array of size 10, using the hash function, $H_1(x) = x \bmod 10$. (***HINT: Please note how EASY modding by 10 is.**)

288, 172, 912, 813, 742, 523

Show the result of the insertions when hash collisions are resolved through:

a) linear probing b) quadratic probing c) separate chaining.

***Note: for part (c), separate chaining, insert at the **END** of the linked list.

a)		b)		c)	
0		0		0	
1		1		1	
2		2		2	
3		3		3	
4		4		4	
5		5		5	
6		6		6	
7		7		7	
8		8		8	
9		9		9	

Extra Page - If you would like any work on this page graded, please label your work clearly.